

# MS&E 228: Applied Causal Inference Powered by ML and AI

## Lecture 18: Dynamic Treatment Effects & Long-Term Effects

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# **Part I: Introduction & Motivation**

From Static to Sequential Interventions

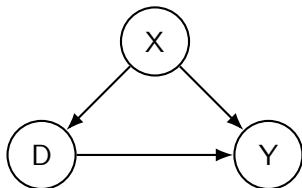
# Goals of this Lecture

Today we move from single-shot interventions to sequences of treatments over time.

1. **Dynamic Treatment Regimes (DTRs):** What are they and why are they a crucial tool for real-world decision-making?
2. **Identification & Estimation:** How can we identify and estimate the mean counterfactual outcome under a specific DTR using the sequential g-formula and debiased machine learning?
3. **Continuous Treatments:** How can we handle continuous treatments in a dynamic setting using a partialling-out approach (Dynamic DML)?
4. **Surrogates:** How can we use short-term surrogate outcomes to estimate long-term treatment effects, and what are the pitfalls?

# From Static to Dynamic Treatments

So far, we have mostly focused on units being treated at **one point in time**, after which we observe an outcome.



But in many settings, units are treated **multiple times over time**. The treatment at one stage can depend on the history of treatments and outcomes up to that point. This is a **Dynamic Treatment Regime**.

# Motivating Example 1: Healthcare

**Scenario:** A cancer patient might receive multiple treatments sequentially (e.g., first-line therapy, then second-line therapy based on response).

Application: Cancer Treatment Regimes (e.g., Talbot, Moodie, & Diorio, 2023)

Cancer patients typically undergo multiple lines of therapy (e.g., chemotherapy, surgery, immunotherapy, radiation therapy, hormonal therapy). These treatments are typically administered in sequence based on prior treatments and on observed response factors / adverse events.

**Causal Question:** What would be the mean counterfactual outcome if all patients first received immunotherapy and then chemotherapy?

## Recent Example: DTRs in Mental Health

**Question:** What is the optimal sequence of treatments for patients with Major Depressive Disorder (MDD) who do not respond to an initial antidepressant?

- ▶ **Domain:** Psychiatry / Depression Treatment
- ▶ **Data Source:** The Sequenced Treatment Alternatives to Relieve Depression (STAR\*D) trial—the largest multi-step clinical trial for depression ( $n = 4,041$ ).
- ▶ **Analysis:** Szmulewicz et al. (2023) emulated a target trial using STAR\*D data to compare different dynamic treatment strategies.

# STAR\*D: Dynamic Treatment Strategies Compared

**Goal.** Evaluate the counterfactual outcome for three treatment strategies for 971 patients failing initial citalopram:

1. **Sequential Monotherapy:** Switch to a different single antidepressant.
2. **Sequential Dual Therapy (SD):** Switch to a combination of two non-SSRI drugs.
3. **Guidelines-Based Strategy (DTR):** Adapts to intermediate response:
  - ▶ *Partial response* → Augment current drug with a second medication.
  - ▶ *No response / intolerance* → Switch to a new antidepressant.

## STAR\*D: Key Results

| Strategy                    | 9-Month Remission | RR vs. Monotherapy      |
|-----------------------------|-------------------|-------------------------|
| Sequential Monotherapy      | 43.5%             | 1.00 (ref)              |
| Sequential Dual Therapy     | 47.6%             | 1.09 (0.90–1.38)        |
| <b>Guidelines-Based DTR</b> | <b>53.2%</b>      | <b>1.22 (0.96–1.46)</b> |

### Takeaway

The **guidelines-based DTR** achieved the highest remission rate and the lowest risk of serious adverse events ( $RR = 0.62$ ). Formalizing clinical guidelines as a DTR and evaluating them with causal methods provided actionable evidence for sequential depression management.



## Motivating Example 2: Marketing

**Scenario:** A customer's journey on a website involves multiple touchpoints. We can show them different ads or offers at different times.

### Application: Customer Lifetime Value

Firms want to optimize a sequence of marketing actions (e.g., ads, discounts) to maximize a customer's long-term value. The decision at each stage depends on the customer's browsing history and prior responses.

**Causal Question:** What is the expected 2-year revenue from a customer if we show them a banner ad on their first visit, and then a 10% discount if they add an item to their cart?

## Motivating Example 3: B2B Operations

**Scenario:** A large cloud provider (like Microsoft Azure) wants to help enterprise customers adopt its services.

Application: Azure Customer Growth (e.g., Battocchi et al., 2021)

Microsoft offers free credits to new customers and can assign a sales specialist to help them. These interventions happen at different stages of the customer lifecycle.

**Causal Question:** What is the expected customer spend if we offer \$10k in initial credits, and then assign a sales specialist if their usage exceeds a certain threshold in the first 6 months?

## Motivating Example 4: Labor Economics

**Scenario:** A government wants to design a program to help unemployed individuals find jobs.

### Real-World Application: Job Training Programs (Bodory et al., 2022)

An individual might first be offered a resume-building workshop. If they are still unemployed after 3 months, they might be offered a more intensive vocational training program.

**Causal Question:** What is the average employment rate after one year if we follow this sequential training strategy?

## Poll 1: Your Experience

Have you ever worked on a problem that involved sequential decision-making or dynamic treatments?

A. Yes, frequently.

B. Yes, once or twice.

C. No, but I see the relevance.

D. Not sure.

# **Part II: A Mathematical Abstraction**

A Two-Period Model of Sequential Treatment

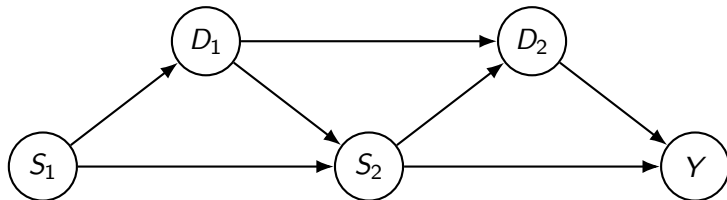
# A Two-Period Model

We can abstract all of these scenarios with the following mathematical notation for a two-period study:

- ▶  $S_1$ : The initial state (pre-treatment covariates) of a unit.
- ▶  $D_1$ : The first period treatment, which may be chosen based on  $S_1$ .
- ▶  $S_2$ : The next period's state. This is affected by both  $S_1$  and  $D_1$ . It can contain qualitatively different variables than  $S_1$ . It can even contain  $S_1, D_1$  as coordinates.
- ▶  $D_2$ : The second period treatment, which may be chosen based on the state  $S_2$ .
- ▶  $Y$ : The final outcome of interest, which is affected by  $S_2$  and  $D_2$ .

# The Causal DAG for a Two-Period Model

The causal relationships between variables in a two-period study:



Note:  $S_2$  is both a **mediator** (on the path  $D_1 \rightarrow S_2 \rightarrow Y$ ) and a **confounder** (common cause of  $D_2$  and  $Y$ ). This dual role is the key challenge.

# The Goal: Estimating Joint Interventions

Our goal is to estimate the mean potential outcome under a **fixed** joint intervention on the treatment sequence.

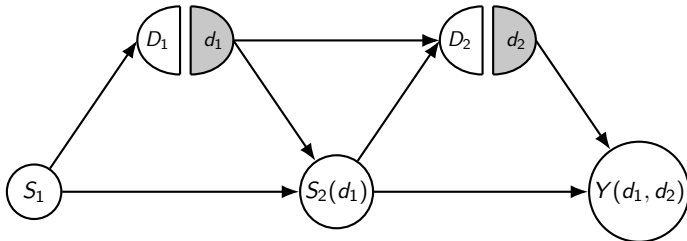
**Estimand:**  $\theta = \mathbb{E}[Y(d_1, d_2)]$

This represents the expected outcome if we were to set the first treatment to  $d_1$  for everyone, and the second treatment to  $d_2$  for everyone.



## The SWIG for a Two-Period Intervention

To reason about the intervention  $\text{fix}(D_1 = d_1, D_2 = d_2)$ , we draw the Single World Intervention Graph (SWIG).



## Revisiting Examples: Healthcare (STAR\*D)

Let's map the STAR\*D trial to our abstraction:

- ▶  $S_1$ : Patient baseline characteristics (age, gender, depression severity).
- ▶  $D_1$ : First-line therapy (e.g., immunotherapy,  $d_1$ ).
- ▶  $S_2$ : Patient status after 3 months (e.g., symptom remission, side effects).
- ▶  $D_2$ : Second-line therapy if no remission (e.g., chemotherapy,  $d_2$ ).
- ▶  $Y$ : Patient outcome at 1 year (e.g., depression-free survival).

**Goal:** Estimate  $\mathbb{E}[Y(\text{immuno}, \text{chemo})]$ .

# Revisiting Examples: Marketing (LTV)

Mapping the customer lifetime value problem:

- ▶  $S_1$ : Customer's initial state (demographics, referral source).
- ▶  $D_1$ : First marketing action (e.g., banner ad,  $d_1$ ).
- ▶  $S_2$ : Customer's state after first action (e.g., pages visited, items in cart).
- ▶  $D_2$ : Second marketing action (e.g., 10% discount,  $d_2$ ).
- ▶  $Y$ : Customer's 2-year revenue.

**Goal:** Estimate  $\mathbb{E}[Y(\text{banner}, \text{discount})]$ .

# **Part III: The Identification Challenge**

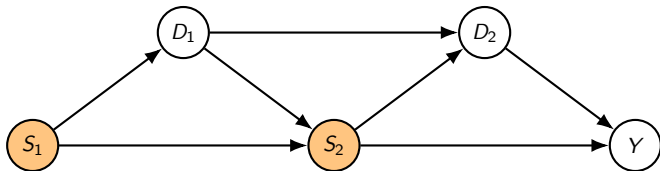
Why Standard Methods Fail

# Why Not Just Condition on All States?

A naive approach: condition on the full history of states  $(S_1, S_2)$ .

$$\mathbb{E}[Y|D_1 = d_1, D_2 = d_2, S_1, S_2] \neq \mathbb{E}[Y(d_1, d_2)|S_1, S_2]$$

**This is incorrect!** Conditioning on  $S_2$  (a post-treatment variable) blocks the effect of  $D_1$  that is mediated through  $S_2$ . It induces collider bias.

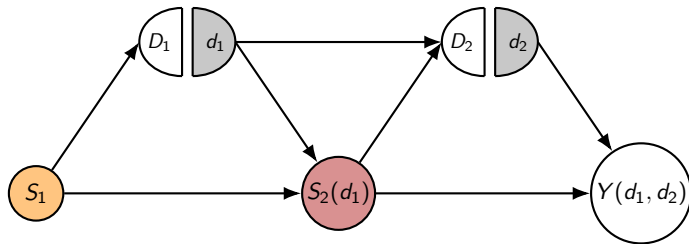


## Why Not Just Condition on All States?

**This is incorrect!** Conditional exogeneity cannot be checked on the SWIG:

$$Y(d_1, d_2) \perp\!\!\!\perp (D_1, D_2) \mid S_1, S_2$$

The variable  $S_2$  does not even appear on the SWIG; only  $S_2(d_1)$ !



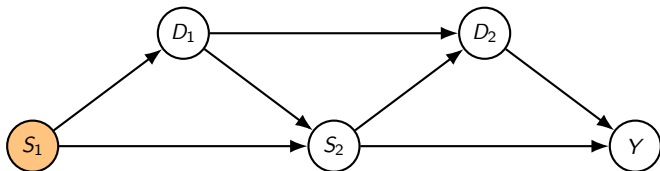
**Mantra:** If you cannot verify it on the SWIG don't use it for identification.

# Why Not Just Condition on Pre-Treatment State?

What if we only condition on the baseline state  $S_1$ ?

$$\mathbb{E}[Y|D_1 = d_1, D_2 = d_2, S_1] \neq \mathbb{E}[Y(d_1, d_2)|S_1]$$

**This is also incorrect!**  $S_2$  is a common cause of  $D_2$  and  $Y$ . Since we haven't conditioned on  $S_2$ , there is an unblocked backdoor path from  $D_2$  to  $Y$  through  $S_2$ , creating confounding.

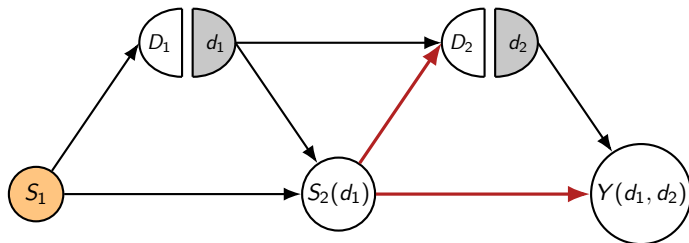


# Why Not Just Condition on Pre-Treatment State?

**This is incorrect!** Conditional exogeneity cannot be verified on the SWIG:

$$Y(d_1, d_2) \perp\!\!\!\perp (D_1, D_2) \mid S_1$$

$d$ -separation does not hold; there is an open path  $D_2 \leftarrow S_2(d_1) \rightarrow Y(d_1, d_2)$ !





## Poll 2: The Main Challenge

What do you think is the fundamental challenge in identifying dynamic treatment effects?

A. Lack of randomization.

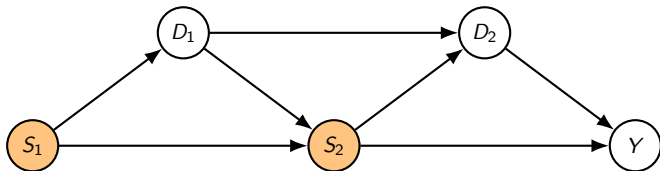
B. Time-varying confounding.

C. Variables that are both mediators and confounders.

D. All of the above.

# What Effects Can we Identify by Conditioning?

The effect on which variable  $V_1$  on what variable  $V_2$  can we identify by conditioning?



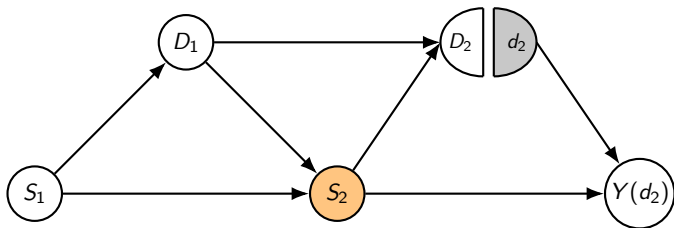
# **Part IV: Identification via Backwards Recursion**

The Sequential G-Formula

## Effects of Just Intervening on Last Period

The effect on only intervening on  $D_2$  on  $Y$  can be identified by conditioning on  $S_2$ .

$$Y(d_2) \perp\!\!\!\perp D_2 \mid S_2$$



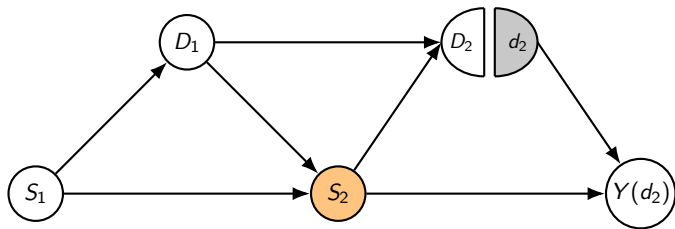
Conditional mean counterfactual outcomes can be identified by conditioning on  $S_2$ :

$$\underbrace{Q_2(S_2, d_2) := \mathbb{E}[Y(d_2) \mid S_2]}_{\text{Forecasted counterfactual outcome based on } S_2 \text{ under second period intervention}} = \underbrace{\mathbb{E}[Y \mid S_2, D_2 = d_2]}_{\text{Forecasted counterfactual outcome based on } S_2 \text{ for samples treated with } D_2 = d_2 \text{ in second period}}$$

## Effect of $D_1$ on $Y(d_2)$ is fully mediated by $S_2$

Assuming we are in a world where we intervene in the second stage.

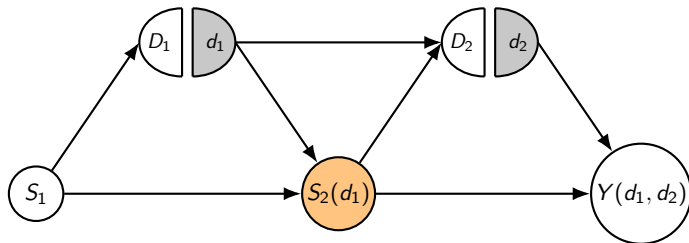
The effect of  $D_1$  on  $Y(d_2)$  is fully mediated by  $S_2$ .



The effect of  $D_1$  on  $Y(d_2)$ , is equal to the effect of  $D_1$  on the “forecasted” counterfactual outcome, forecasted from  $S_2$ , i.e.  $Q_2(S_2, d_2)$ :

$$\underbrace{Q_1(S_1, d_1) := \mathbb{E}[Y(d_1, d_2) \mid S_1]}_{\text{Effect of fix}(D_1 = d_1) \text{ on } Y(d_2)} = \underbrace{\mathbb{E}[Q_2(S_2(d_1), d_2) \mid S_1]}_{\substack{\text{Effect of fix}(D_1 = d_1) \text{ on forecasted } Y(d_2) \\ \text{forecasted from } S_2}}$$

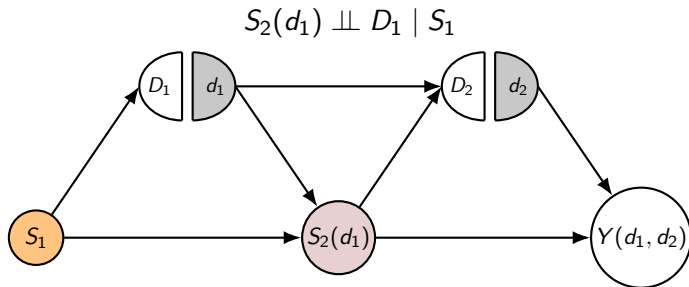
## Formally!



$$\begin{aligned}\mathbb{E}[Y(d_1, d_2) \mid S_1] &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_2(d_1), S_1] \mid S_1] && \text{(tower rule)} \\ &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_2(d_1), D_1 = d_1] \mid S_1] && (d\text{-separation in SWIG}) \\ &= \mathbb{E}[\mathbb{E}[Y(D_1, d_2) \mid S_2(d_1), D_1 = d_1] \mid S_1] && \text{(consistency)} \\ &= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2 = S_2(d_1), D_1 = d_1] \mid S_1] && \text{(consistency)} \\ &= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2 = S_2(d_1)] \mid S_1] && (d\text{-separation in SWIG from prev. slide})\end{aligned}$$

## Effects of Just Intervening on First Period

The effect of only intervening on  $D_1$  on  $S_2$  can be identified by conditioning on  $S_1$ .



The same holds for any function of  $S_2$ , e.g.,  $Q_2(S_2, d_2)$ :

$$Q_2(S_2(d_1), d_2) \perp\!\!\!\perp D_1 \mid S_1$$

Thus we can identify conditional mean counterfactual outcomes by conditioning:

$$Q_1(S_1, d_1) = \mathbb{E}[Y(d_1, d_2) \mid S_1] = \mathbb{E}[Q_2(S_2(d_1), d_2) \mid S_1] = \mathbb{E}[Q_2(S_2, d_2) \mid S_1, D_1 = d_1]$$

## Putting it All Together: Identification via Backwards Recursion

1. Build forecasting model of outcome from second period (state, treatment):

$$Q_2(S_2, D_2) = \mathbb{E}[Y \mid S_2, D_2]$$

2. Construct adjusted outcome for each sample based on  $S_2$ : outcome we expect for the unit if we intervene in second period and fix( $D_2 = d_2$ )

$$Y_{adj} = Q_2(S_2, d_2)$$

3. Build forecasting model of **adjusted outcome** from first period (state,treatment):

$$Q_1(S_1, D_1) = \mathbb{E}[Y_{adj} \mid S_1, D_1]$$

4. Construct adjusted outcome for each sample based on  $S_1$ : outcome we expect for the unit if we **also intervene** in first period and fix( $D_1 = d_1$ )

$$Y_{adj} = Q_1(S_1, d_1)$$

5. Average the final adjusted outcomes:

$$\theta = \mathbb{E}[Y(d_1, d_2)] = \mathbb{E}[Y_{adj}]$$



# The Sequential G-Formula (Robins, 1986)

This backwards recursion leads to the celebrated **sequential  $g$ -formula**

$$\theta := \mathbb{E}[Y(d_1, d_2)] = \mathbb{E}\left[Q_1(S_1, d_1)\right]$$

where  $Q_1$  is defined as a sequence of nested regression models:

$$Q_2(S_2, D_2) = \mathbb{E}[Y | S_2, D_2]$$

$$Q_1(S_1, D_1) = \mathbb{E}[Q_2(S_2, d_2) | S_1, D_1]$$

**Reminder:** one period  $g$ -formula:

$$\theta = \mathbb{E}[Y(d_1)] = \mathbb{E}[g(S_1, d_1)] \quad g(S_1, D_1) = \mathbb{E}[Y | S_1, D_1]$$

## Alternative Derivation (Optional)

$$\begin{aligned}\mathbb{E}[Y(d_1, d_2)] &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_1]] && \text{(Tower Rule)} \\ &= \mathbb{E}[\mathbb{E}[Y(d_1, d_2) \mid S_1, D_1 = d_1]] && (Y(d_1, d_2) \perp\!\!\!\perp D_1 \mid S_1) \\ &= \mathbb{E}[\mathbb{E}[Y(D_1, d_2) \mid S_1, D_1 = d_1]] \\ &= \mathbb{E}[\mathbb{E}[Y(d_2) \mid S_1, D_1 = d_1]] && \text{(consistency)} \\ &= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2, S_1, D_1] \mid S_1, D_1 = d_1]] && \text{(Tower Rule)} \\ &= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2] \mid S_1, D_1 = d_1]] && (Y(d_2) \perp\!\!\!\perp (S_1, D_1) \mid S_2) \\ &= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y(d_2) \mid S_2, D_2 = d_2] \mid S_1, D_1 = d_1]] && (Y(d_2) \perp\!\!\!\perp D_2 \mid S_2) \\ &= \mathbb{E}[\mathbb{E}[\mathbb{E}[Y \mid S_2, D_2 = d_2] \mid S_1, D_1 = d_1]] && \text{(consistency)}\end{aligned}$$

The properties that:

$$Y(d_1, d_2) \perp\!\!\!\perp D_1 \mid S_1 \quad Y(d_2) \perp\!\!\!\perp D_2 \mid S_2$$

are called *sequential conditional ignorability*. The property that  $Y(d_2) \perp\!\!\!\perp (S_1, D_1) \mid S_2$  is expressing that  $S_2$  is a complete mediator of the effect of  $S_1, D_1$  on  $Y(d_2)$ .

# **Part V: Estimation**

From G-Computation to Doubly Robust DTR

# G-Estimation Algorithm

The G-formula gives us a simple plug-in estimation algorithm:

1. Train a model  $\hat{Q}_2$  to predict  $Y$  from  $(S_2, D_2)$ .
2. Predict  $\hat{Y}_{2,i} = \hat{Q}_2(S_{2,i}, d_2)$  for all samples  $i$ .
3. Train a model  $\hat{Q}_1$  to predict these  $\hat{Y}_2$  values from  $S_1$  and  $D_1$ .
4. Predict  $\hat{Y}_{1,i} = \hat{Q}_1(S_{1,i}, d_1)$  for all samples.
5. Estimate  $\theta$  by averaging the final predictions:  $\hat{\theta} = \frac{1}{n} \sum_i \hat{Y}_{1,i}$ .

**Problem:** If we use flexible Machine Learning models for  $\hat{Q}_1, \hat{Q}_2$ , the plug-in estimator  $\hat{\theta}$  will be heavily biased and we can't get valid confidence intervals.

## Debiasing Step 1: Correcting for $Q_1$

The plug-in moment is sensitive to errors in  $Q_1$

$$\theta = \mathbb{E}[Q_1(S_1, d_1)]$$

Add a correction to make it insensitive to  $Q_1$ , which is a regression function:

$$Q_1(S_1, D_1) := \mathbb{E}[Q_2(S_2, d_2) | S_1, D_1]$$

### Step 1: Debiasing $Q_1$ (Doubly Robust Formula)

$$\underbrace{Q_1(S_1, d_1)}_{\text{Original formula}} + \underbrace{\frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 | S_1)}}_{\text{Inverse propensity}} \cdot \underbrace{(Q_2(S_2, d_2) - Q_1(S_1, D_1))}_{\text{Residual of regression defining } Q_1}$$

**Problem:** We are still left with the nuisance  $Q_2$  inside the correction, which we have **not yet debiased**. Formula is still sensitive to  $Q_2$ .

## Debiasing Step 2: Correcting for $Q_2$

We now add a second correction term to debias  $Q_2$ :

### Step 2: Debiasing $f_2$ (Final Doubly Robust Formula)

$$Y_{DR} = Q_1(S_1, d_1) + \frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 | S_1)} (Q_2(S_2, d_2) - Q_1(S_1, D_1))$$
$$+ \underbrace{\frac{\mathbb{I}(D_2 = d_2)}{\Pr(D_2 = d_2 | S_2)}}_{\text{New IPS for period 2}} \cdot \underbrace{\frac{\mathbb{I}(D_1 = d_1)}{\Pr(D_1 = d_1 | S_1)}}_{\text{IPS already multiplying } Q_2} \cdot \underbrace{(Y - Q_2(S_2, D_2))}_{\text{Residual of regression defining } Q_2}$$

# Multiply Robust Formula

The Multiply Robust estimator is defined based on the formula

$$\theta = \mathbb{E}[Y_{DR}] \quad (1)$$

- ▶ We also need to estimate propensity functions  $p_1, p_2$  via ML classification:

$$p_t(S_t) = \Pr(D_t = d_t \mid S_t)$$

- ▶ This formula is insensitive to all nuisance functions (regressions and propensities)

We estimate all these functions in a cross-fitting manner, calculate estimated doubly robust pseudo-outcome  $\hat{Y}_{DR}$  and calculate final estimate

$$\hat{\theta}_{DR} = \mathbb{E}_n[\hat{Y}_{DR}]$$

# Dynamic Doubly Robust Estimator: Pseudocode

```
import numpy as np
from sklearn.model_selection import KFold
# Nuisance models: f2, f1 (regressors), e1, e2 (classifiers)
psi_values = np.zeros(n)
kf = KFold(n_splits=K)
for train_idx, test_idx in kf.split(S1):
    # Stage 2: fit f2(S2, D2) -> Y, and propensity p2(S2) -> D2
    f2_model.fit(np.c_[S2[train_idx], D2[train_idx]], Y[train_idx])
    p2_model.fit(S2[train_idx], D2[train_idx])
    # Stage 1: fit f1(S1, D1) -> f2(S2, d2) and propensity p1(S1) -> D1
    f2_pseudo = f2_model.predict(np.c_[S2[train_idx], np.full(len(train_idx), d2)])
    f1_model.fit(np.c_[S1[train_idx], D1[train_idx]], f2_pseudo)
    p1_model.fit(S1[train_idx], D1[train_idx])

    # Predict on test fold
    f2_hat = f2_model.predict(np.c_[S2[test_idx], D2[test_idx]])
    f1_hat = f1_model.predict(np.c_[S1[test_idx], D1[test_idx]])
    f2_hat_d2 = f2_model.predict(np.c_[S2[test_idx], np.full(len(test_idx), d2)])
    f1_hat_d1 = f1_model.predict(np.c_[S1[test_idx], np.full(len(test_idx), d1)])
    p1_hat = p1_model.predict_proba(S1[test_idx].reshape(-1,1))[:, d1]
    p2_hat = p2_model.predict_proba(np.c_[S1[test_idx], D1[test_idx], S2[test_idx]])[:, d2]

    # Doubly robust score
    w1 = (D1[test_idx]==d1) / np.clip(p1_hat, 0.01, 1)
    w2 = (D2[test_idx]==d2) / np.clip(p2_hat, 0.01, 1)
    psi_values[test_idx] = f1_hat_d1 + w1 * (f2_hat_d2 - f1_hat) + w2 * w1 * (Y[test_idx] - f2_hat)

theta_hat = np.mean(psi_values)
se_hat = np.std(psi_values) / np.sqrt(n)
ci = (theta_hat - 1.96 * se_hat, theta_hat + 1.96 * se_hat)
```



## Main Theorem (Dynamic DR asymptotic normality)

Assume sequential conditional ignorability (implied by the DAG)

$$Y(d_2) \perp\!\!\!\perp D_2 \mid S_2 \quad S_2(d_1) \perp\!\!\!\perp D_1 \mid S_1$$

and  $S_2$  is a complete mediator of the effect of  $D_1, S_1$  on  $Y(d_2)$ , i.e.,  $Y(d_2) \perp\!\!\!\perp (D_1, S_1) \mid S_2$ . Assume **strict** sequential overlap, i.e.,  $\Pr(D_t \mid S_t) \in (\epsilon, 1 - \epsilon)$ . Assume the  $Q$  functions and the propensities are consistent  $\text{RMSE}(\hat{Q}_t), \text{RMSE}(\hat{p}_t) \rightarrow 0$  and the following *Product Rate Conditions* hold:

$$\sqrt{n} \left( \text{RMSE}(\hat{Q}_1) \text{RMSE}(\hat{p}_1) + \text{RMSE}(\hat{Q}_2) \text{RMSE}(\hat{p}_1) + \text{RMSE}(\hat{Q}_2) \text{RMSE}(\hat{p}_2) \right) \rightarrow 0,$$

Under mild regularity conditions, the Dynamic DR Estimator with cross-fitting satisfies

$$\sqrt{n}(\hat{\theta}_{\text{DR}} - \mathbb{E}[Y(d_1, d_2)]) \Rightarrow \mathcal{N}(0, \text{Var}(Y_{\text{DR}})).$$

Standard errors can be estimated as:  $\hat{\text{se}} = \sqrt{\text{Var}_n(\hat{Y}_{\text{DR}})/n}$ , and 95% CI =  $\hat{\theta}_{\text{DR}} \pm 1.96\hat{\text{se}}$ .

# **Part VI: Evaluating Adaptive Policies**

Counterfactual Treatments Can Depend on States

## What if the Policy is Adaptive?

So far, we considered a fixed policy  $(d_1, d_2)$ . What if the policy is **adaptive** (or dynamic), where the treatment decision depends on the current state?

An adaptive policy  $\pi$  is a set of decision rules:

$$\pi = (\pi_1, \pi_2) \quad \text{where} \quad d_1 = \pi_1(S_1) \text{ and } d_2 = \pi_2(S_2)$$

**Example:** In the STAR\*D trial, a policy might be:

- ▶  $\pi_1(S_1)$ : Always start with Citalopram.
- ▶  $\pi_2(S_2)$ : If patient has not remitted ( $S_2$ ), switch to Bupropion; otherwise, continue Citalopram.

Our goal is now to estimate the mean outcome under this adaptive policy:

$$\theta^\pi = \mathbb{E}[Y(\pi_1(S_1), \pi_2(S_2(d_1)))].$$

## Estimation for Adaptive Policies

The G-formula and the doubly robust approach extend naturally.

**G-Computation:** Replace fixed  $d_t$  with the policy function  $\pi_t(S_t)$  at each step.

$$Q_2(S_2, D_2) = \mathbb{E}[Y \mid S_2, D_2]$$

$$Q_1(S_1, D_1) = \mathbb{E}[Q_2(S_2, \pi_2(S_2)) \mid S_1, D_1]$$

### Doubly Robust Formula for Adaptive Policy $\pi$

$$\begin{aligned} Y_{DR} = & Q_1(S_1, \pi_1(S_1)) \\ & + \frac{\mathbb{I}(D_1 = \pi_1(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)} (Q_2(S_2, \pi_2(S_2)) - Q_1(S_1, D_2)) \\ & + \frac{\mathbb{I}(D_1 = \pi_1(S_1))}{\Pr(D_1 = \pi_1(S_1) \mid S_1)} \frac{\mathbb{I}(D_2 = \pi_2(S_2))}{\Pr(D_2 = \pi_2(S_2) \mid S_2)} (Y - Q_2(S_2, D_2)) \end{aligned}$$

## Generalization to $T > 2$ Periods

The entire framework of backwards recursion, G-computation, and doubly robust estimation generalizes to any number of time periods  $T$ .

### G-Formula Recursion:

$$Q_T(S_T, D_T) = \mathbb{E}[Y|S_T, D_T]$$

$$Q_t(S_t, D_t) = \mathbb{E}[Q_{t+1}(S_{t+1}, \pi_{t+1}(S_{t+1}))|S_t, D_t] \quad \text{for } t = 1, \dots, T - 1$$

$$\theta = \mathbb{E}[Q_1(S_1, \pi_1(S_1))]$$

**Doubly Robust Moment:** The moment function will have  $T + 1$  terms, with each term correcting for the estimation error of a nuisance function  $Q_t$  from period  $t = 1, \dots, T$ .

# Part VII: Continuous Treatments

Dynamic Double Machine Learning

## What if Treatments are Continuous?

The inverse propensity weighting approach fails for continuous (or high-dimensional) treatments, as the probability of observing the exact treatment value  $d_t$  is zero.

*What is the analogue of the "Residual-on-Residual" or "Partialling-Out" (FWL) approach for the dynamic treatment regime?*

This leads to an alternative identification and estimation strategy based on so-called **"blip" effects**.

## Identification via "Blip" Effects

Instead of modeling the conditional outcome, we can model the incremental **effect** of each treatment stage.

**Step 2 (Blip):** Define the effect of the second treatment as:

$$\alpha_2(S_2, d_2) := \mathbb{E}[Y(D_1, d_2) - Y(D_1, 0) | S_2]$$

This is the "blip" effect of  $D_2$ . We can identify it by solving a standard single-treatment causal problem, conditioning on  $S_2$ .

**Adjusted Outcome:** We can then create an adjusted outcome that removes the effect of the observed treatment  $D_2$  and adds back the effect of the target treatment  $d_2$ :

$$Y^{\text{adj}} := Y - \alpha_2(S_2, D_2) + \alpha_2(S_2, d_2)$$



# Identification via "Blip" Effects

**Step 1 (Blip):** Define the effect of the first treatment as:

$$\alpha_1(S_1, d_1) := \mathbb{E}[Y(d_1, d_2) - Y(0, d_2) | S_1]$$

This is the "blip" effect of  $D_1$ . We can identify it by solving a standard single-treatment causal problem, conditioning on  $S_1$  on the adjusted outcome from the previous period.

**Adjusted Outcome:** We can then create an adjusted outcome that removes the effect of the observed treatment  $D_1$  and adds back the effect of the target treatment  $d_1$ :

$$Y^{\text{adj}} := Y^{\text{adj}} - \alpha_1(S_1, D_1) + \alpha_1(S_1, d_1)$$

# Dynamic DML Algorithm (Lewis, Syrgkanis (2021))

Assuming linear blip effects:

$$\alpha_t(S_t, d_t) = \delta_t d_t.$$

is the sequential analogue of the Partially Linear Model.

## Dynamic DML (2-Stage, Linear Blips)

1. **Stage 2:** Residualize  $\tilde{Y}_2 = Y - \hat{\mathbb{E}}[Y|S_2]$ ,  $\tilde{D}_2 = D_2 - \hat{\mathbb{E}}[D_2|S_2]$ . Regress  $\tilde{Y}_2$  on  $\tilde{D}_2$  to get  $\hat{\delta}_2$ .
2. Create adjusted outcome:  $Y^{\text{adj}} = Y - \hat{\delta}_2 D_2 + \hat{\delta}_2 d_2$ .
3. **Stage 1:** Residualize  $\tilde{Y}_1 = Y^{\text{adj}} - \hat{\mathbb{E}}[Y^{\text{adj}}|S_1]$ ,  $\tilde{D}_1 = D_1 - \hat{\mathbb{E}}[D_1|S_1]$ . Regress  $\tilde{Y}_1$  on  $\tilde{D}_1$  to get  $\hat{\delta}_1$ .
4. Create adjusted outcome:  $Y^{\text{adj}} = Y^{\text{adj}} - \hat{\delta}_1 D_1 + \hat{\delta}_1 d_1$ .
5. Average the final adjusted outcomes:  $\hat{\tau} = \mathbb{E}_n[Y^{\text{adj}}]$

## Dynamic DML Algorithm as an Insentive Formula

We can view this algorithm as solving the empirical version of the normal equations of the corresponding OLS problems:

$$\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2] = 0$$

$$\mathbb{E}[(\tilde{Y}_1 - \delta_1 \tilde{D}_1) \tilde{D}_1] = 0$$

The target quantity  $\tau = \mathbb{E}[Y(d_1, d_2)]$  is given as a function of  $\delta_1, \delta_2$ :

$$\tau = \mathbb{E}[Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1]$$

## Dynamic DML Algorithm as an Insentive Formula

Note that  $\tilde{Y}_1$  implicitly depends on  $\delta_2$ . We make it explicit:

$$\tilde{Y}_1 = \tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)}, \quad \tilde{V}^{(1)} = V - \mathbb{E}[V \mid S_1]$$

Which gives the system of equations:

$$\begin{aligned} \mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2] &= 0 \\ \mathbb{E}[(\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1) \tilde{D}_1] &= 0 \end{aligned}$$

Both of these equations are insensitive to the nuisance functions, which correspond to the regression problems:  $\mathbb{E}[Y \mid S_2]$ ,  $\mathbb{E}[Y \mid S_1]$ ,  $\mathbb{E}[D_2 \mid S_2]$ ,  $\mathbb{E}[D_2 \mid S_1]$ ,  $\mathbb{E}[D_1 \mid S_1]$ .

## Dynamic DML Algorithm as an Insentive Formula

Parameters  $\theta, \delta_1, \delta_2$  are solution to the following equations, which are insensitive to nuisance functions

$$\begin{aligned}\mathbb{E}[(\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2] &= 0 \\ \mathbb{E}[(\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1) \tilde{D}_1] &= 0 \\ \mathbb{E}[Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1 - \tau] &= 0\end{aligned}$$

The estimated parameters  $\hat{\tau}, \hat{\delta}_1, \hat{\delta}_2$  based on the dynamic DML algorithm are the solution to the empirical analogue of these equations.

# Generic Moment Equation Based Estimator (Chernozhukov et al, 2018)

Suppose that a vector of parameters  $\theta_0$  is the solution to a vector of equations:

$$\theta_0 \text{ is solution to } \mathbb{E}[m(Z; \theta, g_0)] = 0$$

for some known *moment functions*  $m$  that depend on nuisance functions  $g_0$  but are insensitive to the nuisance functions.

Consider the estimated parameters  $\hat{\theta}$  that are the solution to the empirical plug-in analogue of these equations:

$$\hat{\theta} \text{ is solution to } \mathbb{E}_n[m(Z; \theta, \hat{g})] = 0$$

where nuisances  $\hat{g}$  are estimated in a cross-fitting manner.

## Generic Asymptotic Normality (Chernozhukov et al, 2018)

Under regularity conditions and sufficiently fast RMSE rates ( $\ll n^{-1/4}$ ) on the nuisances, the parameters  $\hat{\theta}$  are jointly asymptotically normal:

$$\sqrt{n}(\hat{\theta} - \theta) \rightarrow N(0, J^{-1}\Sigma J^{-\top})$$

where  $J$  is the Jacobian of the equations with respect to the parameters:

$$J = \nabla_{\theta} \mathbb{E}[m(Z; \theta_0, g_0)]$$

and  $\Sigma$  is the covariance matrix of the equations:

$$\Sigma = \mathbb{E}[m(Z; \theta_0, g_0)m(Z; \theta_0, g_0)']$$

Standard errors can be calculated as  $\sqrt{\hat{V}/n}$  using estimated variance  $\hat{V} = \hat{J}^{-1}\hat{\Sigma}\hat{J}^{-\top}$ :

$$\hat{J} = \nabla_{\theta} \mathbb{E}_n[m(Z; \hat{\theta}, \hat{g})] \qquad \hat{\Sigma} = \mathbb{E}_n[m(Z; \hat{\theta}, \hat{g})m(Z; \hat{\theta}, \hat{g})']$$

# Confidence Intervals for Dynamic DML Algorithm

Parameters  $\theta = (\tau, \delta_1, \delta_2)$  are solution to the following equations, which are insensitive to nuisance functions

$$\mathbb{E}[m(Z; \theta, g_0)] = 0 \quad m(Z; \theta, g_0) = \begin{pmatrix} (\tilde{Y}_2 - \delta_2 \tilde{D}_2) \tilde{D}_2 \\ (\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)} - \delta_1 \tilde{D}_1) \tilde{D}_1 \\ Y - \delta_2 D_2 + \delta_2 d_2 - \delta_1 D_1 + \delta_1 d_1 - \tau \end{pmatrix}$$

$g_0$  are the regression functions:  $\mathbb{E}[Y \mid S_2]$ ,  $\mathbb{E}[Y \mid S_1]$ ,  $\mathbb{E}[D_2 \mid S_2]$ ,  $\mathbb{E}[D_2 \mid S_1]$ ,  $\mathbb{E}[D_1 \mid S_1]$

The estimated parameters  $\hat{\tau}, \hat{\delta}_1, \hat{\delta}_2$  based on the dynamic DML algorithm are the solution to the empirical analogue of these equations for all parameters.

**We can apply the general framework to these equations to derive confidence intervals and standard errors.**



# Dynamic DML: Pseudocode — Nuisance Estimation & Closed-Form Solutions

---

```
def dynamic_dml(Y, D1, D2, S1, S2, d1, d2, K=5):
    n = len(Y)
    # Initialize cross-fitted nuisance predictions
    E_Y_S2, E_D2_S2 = np.zeros(n), np.zeros(n)
    E_Y_S1, E_D2_S1, E_D1_S1 = np.zeros(n), np.zeros(n), np.zeros(n)
    kf = KFold(n_splits=K, shuffle=True)
    for train, test in kf.split(S1):
        E_Y_S2[test] = GBR().fit(S2[train], Y[train]).predict(S2[test]) #  $E[Y|S2]$ 
        E_D2_S2[test] = GBR().fit(S2[train], D2[train]).predict(S2[test]) #  $E[D2|S2]$ 
        E_Y_S1[test] = GBR().fit(S1[train], Y[train]).predict(S1[test]) #  $E[Y|S1]$ 
        E_D2_S1[test] = GBR().fit(S1[train], D2[train]).predict(S1[test]) #  $E[D2|S1]$ 
        E_D1_S1[test] = GBR().fit(S1[train], D1[train]).predict(S1[test]) #  $E[D1|S1]$ 
    # Construct residuals:  $\tilde{V} = V - E[V|S_t]$ 
    Y_t2 = Y - E_Y_S2;   D2_t2 = D2 - E_D2_S2           # Stage 2 residuals
    Y_t1 = Y - E_Y_S1;   D2_t1 = D2 - E_D2_S1           # Stage 1 residuals of Y, D2
    D1_t1 = D1 - E_D1_S1                                # Stage 1 residual of D1
    # Closed-form solutions to  $E[m(Z; \theta, g_0)] = 0$ 
    delta2 = np.mean(Y_t2 * D2_t2) / np.mean(D2_t2**2) # OLS:  $\tilde{Y}_2$  on  $\tilde{D}_2$ 
    delta1 = np.mean((Y_t1 - delta2*D2_t1) * D1_t1) / np.mean(D1_t1**2) # OLS:  $\tilde{Y}^{(1)} - \delta_2 \tilde{D}_2^{(1)}$  on  $\tilde{D}_1$ 
    tau = np.mean(Y - delta2*D2 + delta2*d2 - delta1*D1 + delta1*d1) # Sample average
```

---

# Dynamic DML: Pseudocode — Explicit Jacobian & Standard Errors

---

```
# (continued from previous slide)
# Per-observation moment values  $m_i(\theta)$  for  $i=1, \dots, n$ 
m1_i = (Y_t2 - delta2 * D2_t2) * D2_t2 # n-vector
m2_i = (Y_t1 - delta2*D2_t1 - delta1*D1_t1) * D1_t1 # n-vector
m3_i = Y - delta2*D2 + delta2*d2 - delta1*D1 + delta1*d1 - tau # n-vector

# Explicit Jacobian:  $J = E_n[\nabla_{\theta} m(Z; \theta, g_0)]$ 
#  $\theta = (\delta_2, \delta_1, \tau)$ 
#  $dm_1/d(\delta_2) = -E[D2_t2^2]$ ,  $dm_1/d(\delta_1) = 0$ ,  $dm_1/d(\tau) = 0$ 
#  $dm_2/d(\delta_2) = -E[D2_t1 * D1_t1]$ ,  $dm_2/d(\delta_1) = -E[D1_t1^2]$ ,  $dm_2/d(\tau) = 0$ 
#  $dm_3/d(\delta_2) = E[d2 - D2]$ ,  $dm_3/d(\delta_1) = E[d1 - D1]$ ,  $dm_3/d(\tau) = -1$ 
J = np.array([[ -np.mean(D2_t2**2), 0, 0 ],
              [ -np.mean(D2_t1 * D1_t1), -np.mean(D1_t1**2), 0 ],
              [ np.mean(d2 - D2), np.mean(d1 - D1), -1 ]])

# Sandwich variance:  $V = J^{-1} \Sigma J^{-T} / n$ 
M = np.column_stack([m1_i, m2_i, m3_i]) # (n x 3) matrix
Sigma = M.T @ M / n # (3 x 3) covariance of moments
J_inv = np.linalg.inv(J)
V = J_inv @ Sigma @ J_inv.T / n
se_delta2, se_delta1, se_tau = np.sqrt(np.diag(V))
return tau, delta1, delta2, se_tau, se_delta1, se_delta2
```

---

# Takeaways

- ▶ Dynamic Treatment Regimes are a powerful framework for modeling and optimizing sequential decisions in many real-world domains.
- ▶ Simple conditioning fails due to time-varying confounding, where intermediate states are both mediators and confounders.
- ▶ The **sequential G-formula** provides a general identification strategy via backwards recursion.
- ▶ We can construct **doubly robust estimators** for DTRs that allow for flexible ML nuisance models while retaining valid inference.
- ▶ For continuous treatments, **Dynamic DML** provides an alternative estimation strategy based on partialling-out blip effects.

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